Seat No.

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Midterm Examination of Semester 2 — Year: 2024

Subject: 031021112 Electrical Measuring Instruments — Section: 5-6

Date: 10 January 2025 — Time: 10.00-12.00

Name________________________________ ID: ____________________ Class__________

Directions: The test is designed to measure your comprehension. The test is divided into 17 questions. There will be 9 pages (including this page) and they are worth 70 points.

- This exam paper contains no errors. If a suspected error is found, it is the student's discretion to correct it.
- Answer the questions on this test paper.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators can be used in this test.

Now begin the test.

วิทยาลัยเทคโนโลยีอุตสาหกรรม

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

This exam paper is part of a set used to assess student abilities as follows:

		Q1-10	Q11-13	Q14-16
CLO 1	Describe the basic knowledge about principles and operation of electrical instruments and measurements.	√		
CLO 2	Explain the major categories of error and calculate the percentage error including the loading effect.		√	
CLO 3	Explain and design the electrical quantity measurements by using Voltmeter, Ammeter and Ohmmeter with extending scales.			√

Q1) Define the following terms: (4 marks)

(a) Measurement :..

(b) Accuracy :...

(c) Precision :...

d) Gross Error: ....

Q2) In calculating voltage drop, a current of **2.15 A** is recorded in a resistance of **25.7 Ω**. Calculate the voltage drop across the resistor with ***four significant figures***.

(2 marks)

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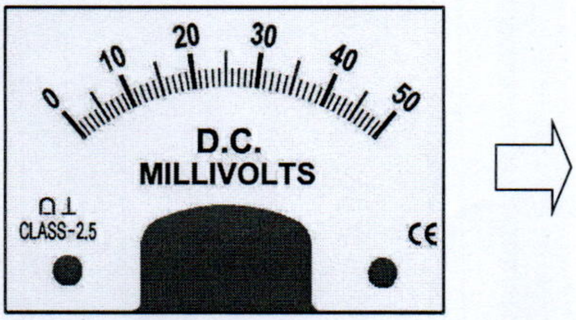
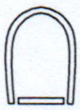
Q3) Identify each of the instrument symbols illustrated in Figure (a)–(b). (2 marks)





a) ....

Class 2.5

b) ....

Q4) Convert the following numbers as the required unit. (2marks)

a) Convert 100,000 Ω to megohms ....

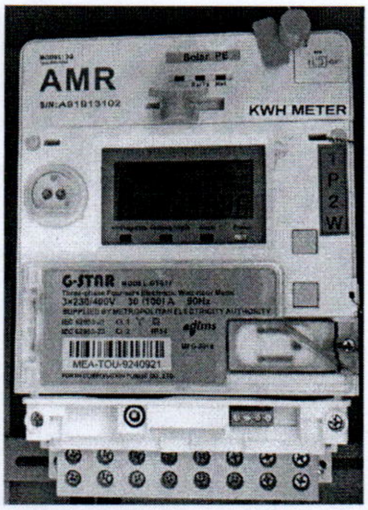
b) Convert 0.0017 A to milliamperes ....

c) Convert 47000 pF to microfarads ....

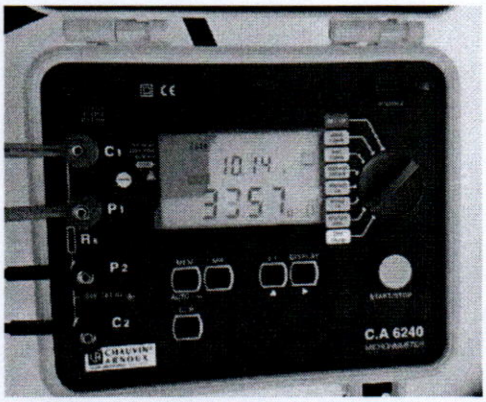
d) Convert 0.0075 s to nanoseconds ....

Q5) Describe the basic function and the benefit of Time of Use Meter (TOU Meter).

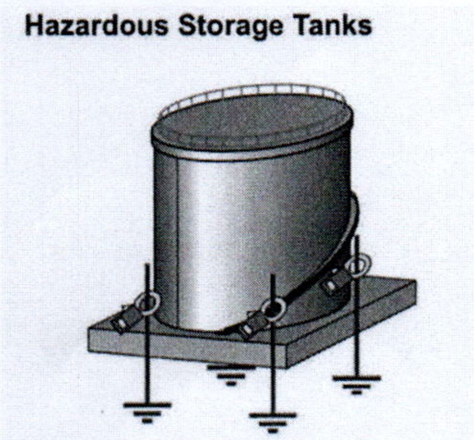
(2 marks)



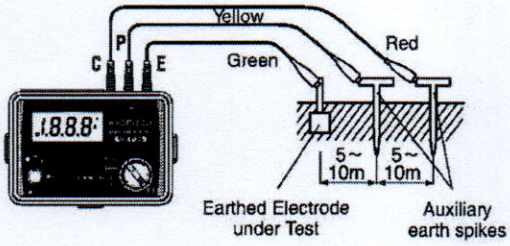
Q6) What is the application of Microhmmeter? (2 marks)



Q7) Which is the proper instrument that we can use to measure Multiple Grounding of Hazardous Storage Tanks? (2 marks)



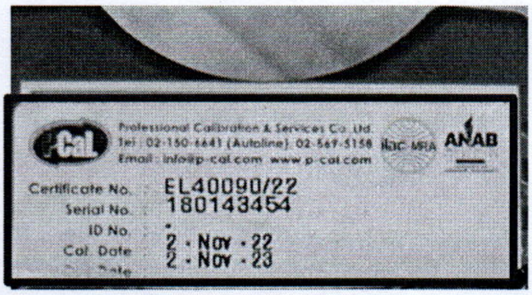
Q8) What is the standard value of Earthed (Grounded) Electrode testing resistance (Ω) for the grounding system for residence in Thailand? (2 marks)



Earth Electrode testing resistance

$\leq$ .... Ω

Q9) Please explain the meaning of the instrument calibration as shown in the Figure. (2 marks)



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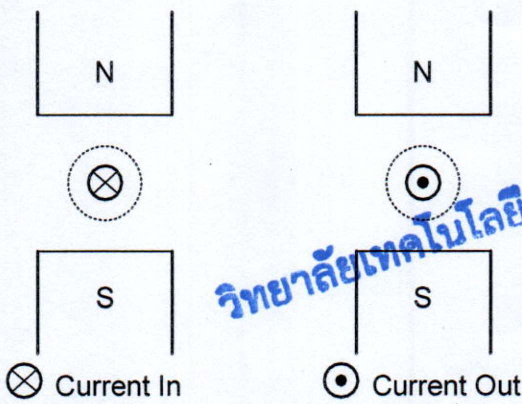
Q10) Using ***Right Hand Grip Rule*** to sketch the direction of the ***Magnetic field*** and *Force*. (2 marks)



Q11) List and explain the **three forces** involving in the moving system of a deflection instrument. (3 marks)

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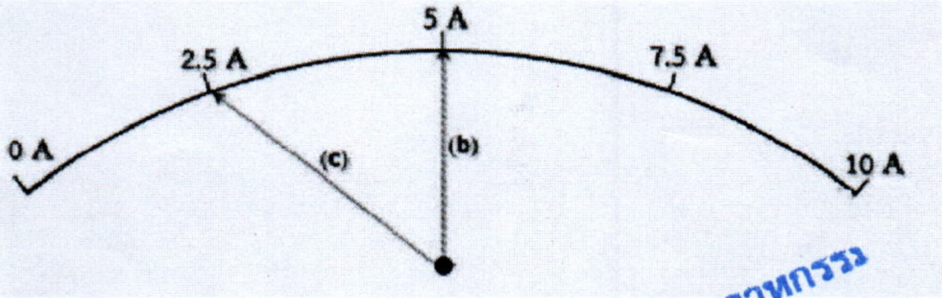
Q12) The expected value of a resistor is 10 kΩ, but the measurement using ohmmeter yields a value of 9.95 kΩ. Calculate **(a) absolute error, (b) % error** and **(c) % of accuracy**. (3 marks)

Q13) An Ammeter range 10A class 1.5, calculate the maximum %Error for 2 different position of **Case (b)** = 5A, **Case (c)** = 2.5A (6 marks)

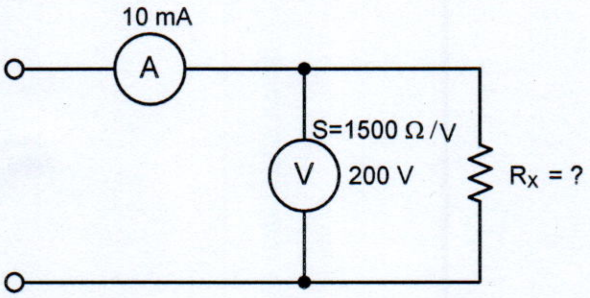


Q14) A voltmeter having a sensitivity of 1,500 Ω/V, reads 200 V on its 250 V scale when connected across an unknown resistor (Rx) in series with an ammeter. When the ammeter reads 10 mA, Calculate the following items:

(a) The apparent resistance of the unknown resistor (RT),

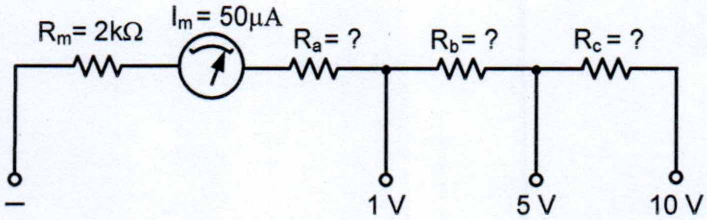
(b) The actual resistance of the unknown resistor (RX),

(c) % error due to the loading effect. (6 marks)

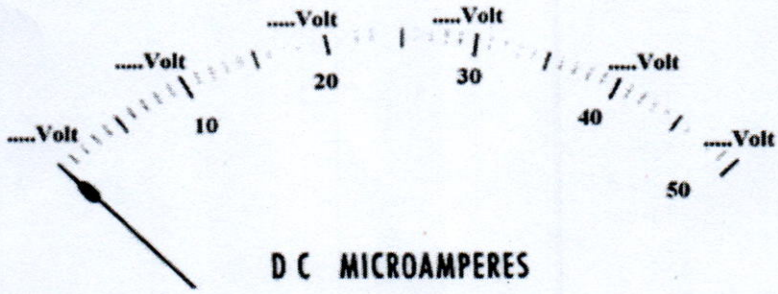


Q15) A PMMC instrument with FSD = 50 μA and Rm = 2 kΩ, is to be employed as a volt-meter with ranges of 1 V, 5 V and 10 V. (10 marks)

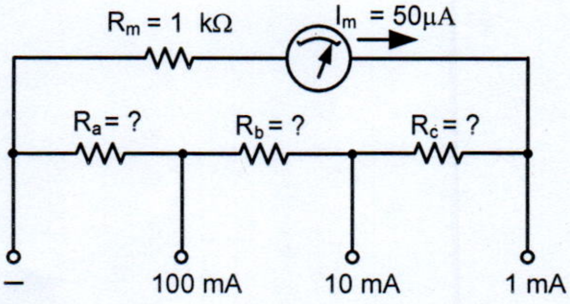
a) Calculate the required values of multi-plier resistors (Ra, Rb, Rc).



b) Determine how the voltmeter scale should be marked for the **Range 10V.**



Q16) A PMMC instrument has 50 μA and a coil resistance of 1 kΩ. Calculate the required resistance (Ra, Rb, Rc) values to convert the instrument into an ammeter (multi-range) with an FSD of 100 mA, 10 mA, and 1 mA (10 marks)



Q17) The series ohmmeter in figure is made up of a 1.5 V battery, a 50 μA meter, $R_m = 2\ \text{k}\Omega$. (10 marks)

(a) Calculate the value of R_1 to make the current to be full scale.

(b) Determine how the resistance scale should be marked at $R_x = 0$ and $R_x = \infty$.

(c) Determine how the resistance scale should be marked at 0.75 FSD, 0.5 FSD, and 0.25 FSD.